A Post-Fabrication-Tunable Time-Domain Multi-Row Read for 1T1R RRAM Processing-in-Memory in 130 nm SG13G2

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***Abstract*—A research chip's RRAM is characterized only after tape-out. A processing-in-memory (PiM) read designed for one operating point can therefore miss it. This paper presents a read path whose operating point is chosen after fabrication, and a method that turns its simulation into a budget for the device. The read path is part of a tape-out-ready 1T1R macro in IHP's 130 nm SG13G2 process. Each column converts the bit-line discharge time into a thermometer code with a delay-line TDC, so opening several rows turns a read into an in-memory NOR or majority. A controller sets the converter's window, tap pitch and drive levels after fabrication. Transistor-level simulation with the extracted bit-line loading and the RRAM device capacitance gives each operation's margin, and a Monte Carlo converts it into the resistance spread it tolerates. The budget sets the device requirement: the single-row read works at IHP's median high-resistance state, and multi-row logic requires the deep-reset state with a verify-tight spread, both within the module's programming options. All results are pre-silicon.**

***Index Terms*—RRAM, 1T1R, processing-in-memory, time-domain readout, time-to-digital converter, IHP SG13G2.**

I. Introduction

Resistive RAM (RRAM) in a one-transistor-one-resistor (1T1R) array stores data as conductance. When several word lines open at once, the bit line sums the conductances of the selected cells, and the array becomes a computing element. A multi-row read then yields bulk bitwise operations [1] or an analog multiply-accumulate [2], [3] in a single access [4]. Stateful memristive logic complements this by writing the result into a cell: material implication [5], NOR [6], majority [7]. Silicon demonstrations of RRAM processing-in-memory (PiM) rely on dedicated sense circuits with tracking references or calibration [8], [9], and time-domain readout with TDCs digitizes in-memory multiply-accumulate compactly [10], [11]. On an academic research chip, however, the operating point of the memory element is known only after tape-out, and so are its resistance spread and its parasitic capacitances. The readout therefore has to be *tunable after fabrication*, not optimized for one assumed point.

This paper presents a PiM read path designed for that situation, verified to a tape-out-ready 64×64 macro in the IHP SG13G2 130 nm BiCMOS process with its back-end HfAlO RRAM module [12]. Instead of voltage-mode sense amplifiers, every column converts the bit-line *discharge time* into an 8-bit thermometer code with an 8-tap delay-line TDC. With several rows open the discharge time measures the summed cell conductance, i.e. an in-memory NOR or majority. The sampling instant, tap pitch and drive levels are all set from software, through dedicated supply pads and a register-programmable controller. Unlike the reference-tracked or calibrated sense circuits of [8], [9] and the MAC-oriented TDC readouts of [10], [11], this is a bitwise-logic read whose decision window is placed after fabrication. All results are pre-silicon.

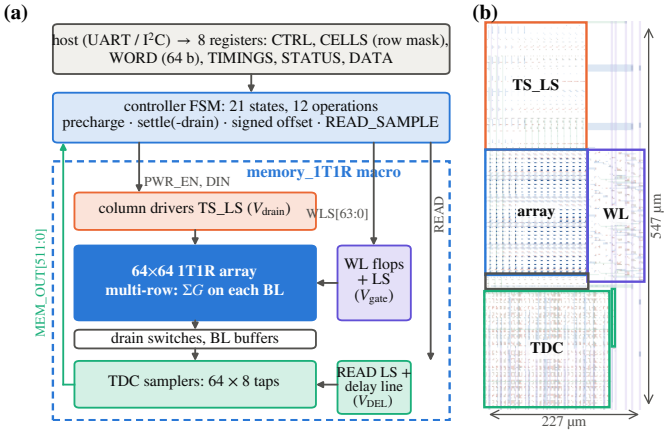


Fig. 1. (a) Macro and controller architecture: the host writes the row mask, data word and timings; the FSM sequences the drivers, word lines, drain switches and READ, and the 512-bit thermometer result returns through the DATA register. (b) The tape-out-ready layout, 227×547 µm, blocks outlined.

The contributions are: (i) a time-domain multi-row read whose window position, tap pitch and drive levels are set after fabrication by a 21-state controller, co-verified in SPICE/AMS with the RTL (Sections II, III); (ii) a transistor-level evaluation of NOR and majority as decision margins in TDC taps over the read supplies, the state resistance and the number of rows, with energy and read disturb (Section IV); and (iii) a budget derived from those margins: the resistance spread, corners and temperatures each operation tolerates, i.e. what the RRAM cell has to provide for the read to work.

II. Macro and Time-Domain Multi-Row Read

A. Array and drivers

The macro (226.5×547.4 µm², Fig. 1, Table III) holds 4096 1T1R cells on a 2.26×2.77 µm pitch, each an HV nMOS (W/L

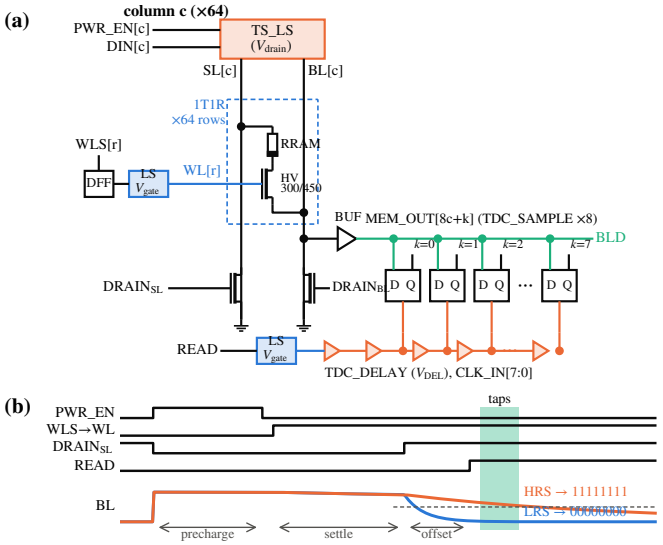


Fig. 2. (a) One of the 64 columns of the macro (from the layout netlist). The RRAM top electrode connects to SL, the high-voltage (HV) access transistor to BL. (b) Read sequence: precharge BL to V_{drain}, open the selected word lines, then discharge through DRAIN$_{SL}$; the TDC samples BL at eight taps after the READ edge.

= 300/450 nm) in series with an IHP MEMRES device whose top electrode connects to the source line SL[c] and whose access transistor connects to the bit line BL[c] (Fig. 2(a)). Each column has a tri-state level-shifting driver (TS_LS) from V_{drain} that drives SL or BL high (SET or RESET polarity, the latter also the read precharge) or leaves both floating. Two global HV switches, DRAIN$_{SL}$ and DRAIN$_{BL}$, ground all source or bit lines. Each word line is driven through a pipeline flip-flop and a cross-coupled level shifter (LS) supplied from V_{gate}, so a row-select mask written in one cycle opens the word lines in the next. An arbitrary 64-bit row mask opens any subset of rows at once, which turns a read into a PiM operation.

B. Time-domain readout

A read precharges BL to V_{drain}, opens the selected rows and then turns on DRAIN$_{SL}$. BL discharges through the selected cells with a time constant set by the cell resistance, the access-transistor resistance and the bit-line capacitance. A 1.2 V buffer with a threshold of about 0.6 V converts BL into a digital signal BLD, and eight flip-flops sample BLD on the eight taps CLK_IN[7:0] of a delay line triggered by READ (Fig. 2(a)). The line is shared by all columns and supplied from its own pad V_{DEL}, which sets the tap pitch. The result, MEM_OUT[8c+k], is a thermometer code that counts the taps at which BL was still high: a low-resistance (LRS) cell gives a short code and a high-resistance (HRS) cell a long one (Fig. 2(b)).

With n rows open, the column conductance is the sum of the selected cells, so the discharge time, and with it the code $c(k)$, falls monotonically with the number k of LRS cells. The in-memory operations are code thresholds, each with a margin measured in taps: NOR over the n rows separates $k = 0$ from $k \geq 1$, margin $c(0) - c(1)$; majority separates $k \leq n/2$ from $k > n/2$, margin $c(n/2) - c(n/2+1)$.[1] Three knobs place the decision: the TDC fires a fixed delay after READ, so the controller's offset between DRAIN$_{SL}$ and READ moves the sampling instant in 10 ns steps; V_{DEL} sets the tap pitch; and V_{gate} sets both the level-shifter delay and the access-transistor resistance.

III. Controller and Read Sequencing

Each macro is driven by a 21-state FSM with 12 operations (Fig. 1): single and complete reads, SET, RESET and mixed-polarity writes of a 64-bit word into the selected rows, precharge, a discharge-length test (READ pulsed with the drain on until the column code reaches zero, measuring slow or pristine cells) and a sample capture (READ_SAMPLE: a READ pulse every 20 ns, sixteen codes). A read precharges for a programmable number of cycles, opens the row mask for a settle time, and applies DRAIN$_{SL}$ and READ with a *signed* offset: a positive offset turns the drain on first, so the TDC samples later in the discharge. Eight 32-bit registers (control, row mask, data word, timings, status, paged result) are written from a host over a serial link, and a host procedure searches V_{gate}, V_{DEL}, V_{drain}, settle and offset for the operating point that best separates the classes of each operation, the procedure used in Section IV.

IV. Pre-Silicon Evaluation of the PiM Read

A. Setup

An 8×8 slice of the macro's layout netlist, with its drivers, word-line flops and level shifters, drain switches, delay line and TDC flops, is simulated with the foundry PSP models (typical corner). Cells are linear resistors: 9 kΩ for LRS (IHP's LRS2 target) and, for HRS, 200 kΩ, the top of IHP's characterized range (1 µA at 0.2 V); 1 MΩ is added as an *ideal case*. At the 0.06–0.12 V read bias the HRS non-linearity of these devices raises R by about 10 % over its 0.2 V value and the LRS is ohmic [13]; IHP's compact model, at the two states it can produce (1.4 and 12 kΩ), reproduces the resistor runs' crossing times within 0.06 ns at the three Table I settings. The supplies are confined to $V_{drain}, V_{gate}, V_{DEL} \in \{0.8, 0.9, 1.0\}$ V so that a read never approaches a write condition, and the voltage across the selected cells and their current are recorded in every run. An access takes the precharge (below), the settle, the offset and the READ pulse: about 0.45 µs for 512 result bits at 100 MHz, mostly precharge, and 8–13 pJ per 8×8 slice from the supply currents (65–105 pJ per access scaled to the macro, controller excluded). 90 % of it is the 1.2 V clock energy of the 64 word-line flip-flops, which toggle every cycle during the 40-cycle precharge that the extracted loading needs at $V_{drain} = 0.8$ V; the read path itself is below 0.3 pJ.

Capacitance model. A resistor cell ignores two capacitances that set the discharge. The first is the line loading: a Quantus RC extraction of the macro (21 930 MOS and 4096 MEMRES)

[1] A population count (a distinct code per k) is obtained only for two rows (Section IV-C).

TABLE I
BEST OPERATING POINT PER OPERATION WITHIN 0.8–1.0 V (SLICE, LRS 9 kΩ)

op.	HRS (Ω)	$V_{drain}/V_{gate}/V_{DEL}$ (V)	off. (ns)	codes $c(k)$ $k = 0 \ldots 8$	margin (taps)	V_{RRAM}/I_{cell} (V/µA)
read, 1 row	200 k	0.8/0.8/0.8	20	8, 0 (HRS, LRS)	8	0.09/1.3
NOR, 8 rows	200 k	0.9/0.8/1.0	10	7 1 0 0 0 0 0 0 0 0	6	0.06/0.8
majority-8	200 k	1.0/0.8/1.0	10	8 8 8 6 3 1 0 0 0	2	0.06/0.8
NOR, 8 rows	1 M†	0.8/0.8/0.9	10	8 0 0 0 0 0 0 0 0	8	0.12/1.0
majority-8	1 M†	1.0/0.8/1.0	10	8 8 8 8 7 3 0 0 0	4	0.12/1.0

direct transients, C_{MIM} = 10 fF, extracted line capacitance; †ideal HRS

gives 47–62 fF of interconnect on each BL and 21–36 fF on each SL; the slice adds 70 fF to BL (wire plus the source junctions of the 56 rows it does not instantiate) and 30 fF to SL. The second is the MEMRES device itself, a metal-insulator-metal stack that neither IHP's compact model (whose only state is the filament gap) nor the extraction contains: for its design-rule area of 0.36 µm², the model's 6 nm HfAlO and ε_r = 12–20, a Sentaurus solution of the drawn stack (Metal2 plate, Via2, Metal3) gives C_{MIM} = 6.8–11.0 fF per cell, 6 % above the parallel plate, which bypasses a selected cell with a time constant of 2 ns at 200 kΩ and 10 ns at 1 MΩ, the scale of the differences the read resolves. All results use C_{MIM} = 10 fF ; the line loading, which doubles every crossing time, also makes the default 10-cycle precharge leave BL at 0.5 V for V_{drain} = 0.8 V, so all runs precharge for 40 cycles (400 ns).

B. Single-row read

The macro's Liberty view assumes V_{gate} = 4 V, where the cross-coupled level shifters take 89–126 ns and the RTL controller in AMS never captured a sample; at 0.8–2.0 V they settle within 3.7 ns, so the controller was re-sequenced to open the word lines before the read. At V_{gate} = 0.8 V the access transistor is in weak inversion and limits the cell current to 0.5–1.7 µA (200 to 9 kΩ, 0.9 V on BL), so the 22:1 ratio of the cells reaches the bit line as a 3.6:1 current contrast. A single row then separates HRS from LRS by the full eight taps: the 200 kΩ cell crosses the buffer threshold 50.7 ns after the drain opens and the LRS cell 28.2 ns ($V_{drain} = V_{gate} = V_{DEL}$ = 0.8 V), a window at +20 ns lies between them, and the cell sees 0.09 V and 1.3 µA.

C. Multi-row read: NOR and majority

The PiM read opens n rows of one slice column in a single access with k cells in LRS and $n - k$ in HRS. At $V_{drain} \leq 1$ V the settle cycle must ground SL (the controller's settle-drain bit): with SL floating, charge sharing takes BL below the buffer threshold before READ. The discharge starts when the word lines open, and the eight-tap window is placed on it by the signed read offset in 10 ns steps. The full grid $V_{drain}, V_{gate}, V_{DEL} \in \{0.8, 0.9, 1.0\}$ V was simulated for HRS 200 kΩ and 1 MΩ; codes at every window position follow from the simulated crossing times and the measured tap tables, with the comparison offset fitted on direct transients, and the best point of each operation was then simulated directly (Table I).

TABLE II
MARGIN (TAPS) VS. TEMPERATURE, TT, 200 kΩ: TABLE I SETTING FIXED AT 27 °C / RE-TUNED (V_{drain}, V_{gate}, V_{DEL}, OFFSET) PER TEMPERATURE

	−40 °C	27 °C	125 °C
read, 1 row	8 / 8	8 / 8	8 / 8
NOR, 8 rows	0 / 5	5 / 5	0 / 1
majority, 8 rows	0 / 2	2 / 2	0 / 0

Where the window lands. At 200 kΩ every multi-row operation is resolved at V_{gate} = 0.8 V and at no higher word-line voltage (Fig. 3(c),(d)): the current-limited cells stretch the 8-row discharge to 20–27 ns (Fig. 3(a)) and V_{drain} decides where the band falls against the window grid; at $V_{gate} \geq 0.9$ V the band is shorter than the 10 ns step and falls between window positions. That step, one clock period, is the design's main limitation: it caps the row count and the hot-corner margins below, and a fractional-cycle offset is the first change to make. Over the PSP corners and −40 to 125 °C the 8-row band moves from 16 to 60 ns against the 10 ns window step; re-tuning V_{drain}, V_{gate}, V_{DEL} and the offset per temperature (Table II) keeps the single-row read at eight taps and recovers the 8-row NOR at −40 °C (five taps) but not at 125 °C (one tap), where the band is shorter than the step.

NOR. At V_{drain} = 0.9 V the all-HRS crossing (23.9 ns) lies inside the window and the $k = 1$ crossing (22.9 ns) before it: codes 7 1 0 0 0 0 0 0 0, a NOR by six taps; the ideal 1 MΩ reaches the full scale.

Majority. The $k \to k+1$ step scales as $1/k^2$, so $c(4) - c(5)$ needs the window on the compressed part of the band: V_{drain} = 1.0 V moves it there, codes 8 8 8 6 3 1 0 0 0, a majority by two taps (four at 1 MΩ).

Rows per access. The same model was run with 1 to 8 rows open over the full supply grid, and with 16 to 64 rows at the best 8-row supplies (Fig. 4). At 200 kΩ the NOR keeps all eight taps up to four rows and four to five taps up to eight; the majority falls from eight taps at two rows to two or three at six to eight. Every open row adds an HRS current that one LRS cell must stand out against, so the crossing-time step behind each decision falls roughly as $1/n^2$ (Fig. 4(b)), from 17 ns (NOR) and 5 ns (majority) at two rows to 1.4 and 0.4 ns at eight. From 16 rows on the 200 kΩ step is at or below a tap pitch, and single-tap margins computed there are tap-edge artefacts. With the ideal 1 MΩ the NOR step is still 2.0 ns at 16 rows, but the window, 1.2–2.5 ns wide and placed in 10 ns steps, misses it by 0.03 ns; at IHP's HRS one access decides over at most eight rows, and wider sums are split into eight-row groups combined by the host. Only two rows give a distinct code for every count.

Read disturb and spread. A read biases the selected cells in the RESET polarity (BL high, SL grounded), so repeated reads push toward HRS. V_{gate}, not V_{drain}, sets the stress: at the best points a cell sees at most 0.06–0.12 V and 0.8–1.3 µA, below IHP's 0.2 V read condition and a sixth of the 8 µA RESET target. On these devices 10^5 pulses of 10 µs at 0.1–0.2 V (SET polarity) leave every level unchanged and drift starts at 0.3 V,

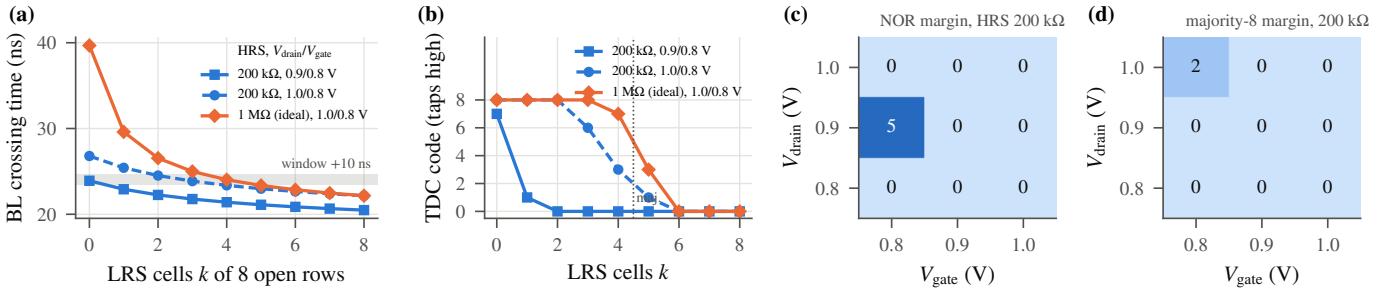


Fig. 3. Multi-row read with the extracted line capacitance and C_{MIM} = 10 fF, 8 rows open, k cells at 9 kΩ and 8 − k at the HRS value, 40-cycle precharge, settle-drain, V_{gate} = 0.8 V. (a) BL crossing time vs. k at the best points; shaded: the TDC window at the +10 ns read offset (first READ_SAMPLE pulse), V_{DEL} = 1.0 V. (b) Simulated codes at those points. (c), (d) Best NOR and majority margin over $V_{drain} \times V_{gate}$ for HRS 200 kΩ, at the best V_{DEL} and window position, computed from the grid's crossing times.

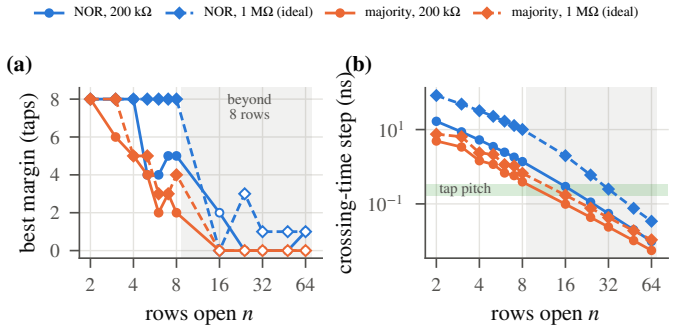


Fig. 4. Rows per access, V_{gate} = 0.8 V. (a) Best NOR and majority margin vs. rows open, over V_{drain}, V_{DEL} and window position (1–8 rows: full supply grid; hollow, 16–64 rows: at the best 8-row supplies). (b) Crossing-time step behind each decision, $t(0) - t(1)$ for NOR and $t(n/2) - t(n/2+1)$ for majority; shaded: the tap pitch at V_{DEL} = 0.8–1.0 V.

while retention alone lowers the LRS by 15 % in 100 h at room temperature [14]: the refresh interval is set by retention and temperature rather than by the read count. The margins above are at the typical corner with fixed resistances. With a 3–3.6:1 current contrast, a Monte Carlo on the simulated crossing times, checked against 160 SPICE runs with drawn cell resistances, limits the HRS spread for an n-row decision at 10^{-3} error to $\sigma_{\ln R} \approx 0.46/\sqrt{n}$ with an ideal threshold, 0.16 at eight rows and 0.33 at two, against 0.3–0.6 measured on IHP's 250 nm arrays and 0.5–1 on the Al-doped 130 nm devices, whose HRS at the same verify has a median of 90–220 kΩ but a conductive edge pinned at 50 kΩ [12], [15]; the single-row read tolerates 0.45. A fivefold deeper reset, the 1 MΩ column of Table I, buys a full-scale 8-row NOR, a four-tap majority and an eight-row spread limit of 0.5, inside the measured range. A deeper reset, a tighter verify window and fewer rows per access are the levers; a higher word line adds only 30–45 % of contrast. Both are programming-time costs: a deeper reset takes more ISPVA pulses per write (about twelve of 12 µs at the 5 µA target [12]) and, with Al doping, leaves a quarter of the cells at the read floor [15]; a tighter window costs further verify iterations and the cells that never enter it.

TABLE III
MACRO AND CONTROLLER SUMMARY

Technology	IHP SG13G2 130 nm, BEOL HfAlO MEMRES
Array	64×64 1T1R, HV 300/450 nm, pitch 2.26×2.77 µm
Macro	226.5×547.4 µm²; 4096 MEMRES, 21 930 MOS
Area split	array 21 %, col. drivers 21 %, WL 12 %, TDC 23 %
Controller	21-state FSM, 12 ops, 1673 cells (0.025 mm²)
Signoff	100 MHz (slow corner); DRC, antenna and LVS clean
Energy	65–105 pJ per 512-bit access (0.13–0.2 pJ/bit), slice SPICE

TABLE IV
COMPARISON WITH RRAM AND TIME-DOMAIN CIM MACROS

	node	memory	readout	summed[a]	reported metric[f]
[8]	65 nm	1 Mb 1T1R	current SA, 3 b	9–25	n/r, 14.8 ns/cycle
[9]	130 nm	3 M 1T1R	neuron ADC[b]	256	43 TOPS/W (1 b)
[10]	40 nm	delay lines only	counter TDC	25–150, serial	12.1 TOPS/W[c]
[11]	180 nm	8 kb SRAM	counter TDC	32 per chain	0.3–0.8 TOPS/W[d]
this work[e]	130 nm	4 kb 1T1R	8-tap TDC	≤ 8	0.13–0.2 pJ/bit

All but this work are measured silicon MAC/MVM macros. [a]cells sensed in one access. [b]per-layer calibrated. [c]compute only, no memory on chip. [d]8×8 b at 1.2 V; 768–2124 TOPS/W is normalized to 28 nm, 1 b. [e]tape-out-ready layout, slice SPICE, 0.45 µs per access. [f]each work's own figure; MAC metrics are not comparable with a bitwise read, the table compares structure.

V. Implementation and Comparison

The macro is full custom, the controller synthesized and placed around it (Table III). Against the MAC macros of Table IV, with their reference-tracked or calibrated readouts, this work trades throughput for a bitwise read whose every timing and drive level is a register or a pad.

VI. Conclusion

A post-fabrication-tunable time-domain read for 1T1R RRAM PiM has been co-verified to a tape-out-ready 64×64 macro on IHP SG13G2. Its simulated budget is a design requirement, not a verdict: the single-row read works at IHP's median HRS, and 8-row NOR and majority require the deep-reset state with a spread of 0.16 (eight rows) or 0.33 (two), which a deeper reset or a tighter verify provides. Silicon must first measure that spread.

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